**Stochastic Processes 2021/22 Semester 1**

**Tutorial 8 Extension Questions**

*Please note: These questions are an extension to Tutorial 8 for individual training only and are not supposed to be discussed during tutorials or office hours. Answers are available for all questions, but worked solutions are only available for some hard questions.*

1. Answer the questions (a) to (c) in Tutorial 8 Question 3 with the new transition rate matrix:

$$Q = \begin{pmatrix} -3 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 1 & -4 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 & 2 \\ 1 & 1 & 1 & -5 & 1 & 1 \\ 0 & 0 & 0 & 3 & 0 & -3 \end{pmatrix}$$

2. Answer the questions (a) to (c) in Tutorial 8 Question 3 with the new transition rate matrix:

$$Q = \begin{pmatrix} -2 & 1 & 1 & 0 & 0 & 0 \\ 1 & -3 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

3. Redo Tutorial 8 Question 4 with the new matrix of transition rates (generator)

$$Q = \begin{pmatrix} -7 & 3 & 4 \\ 1 & -2 & 1 \\ 3 & 2 & -5 \end{pmatrix}$$

4. Redo Tutorial 8 Question 4 with the new matrix of transition rates (generator)

$$Q = \begin{pmatrix} -5 & 2 & 3 \\ 4 & -6 & 2 \\ 1 & 5 & -6 \end{pmatrix}$$